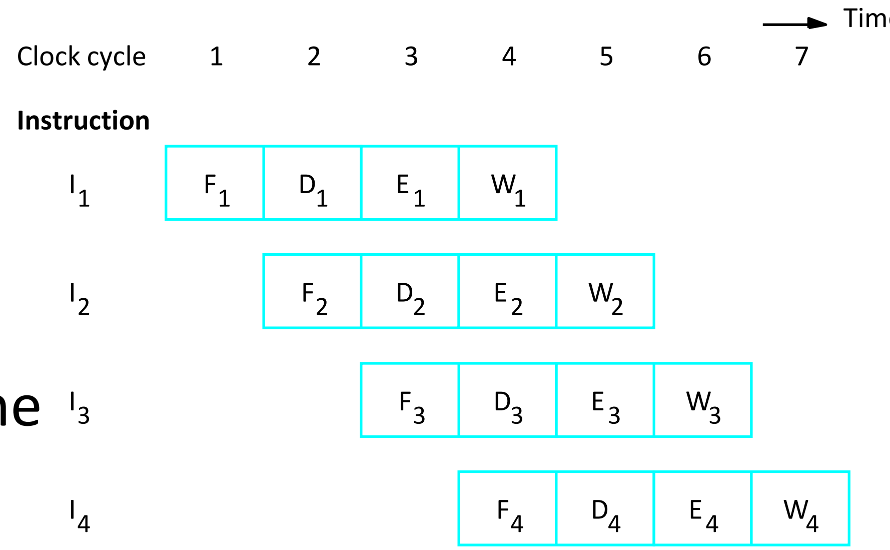
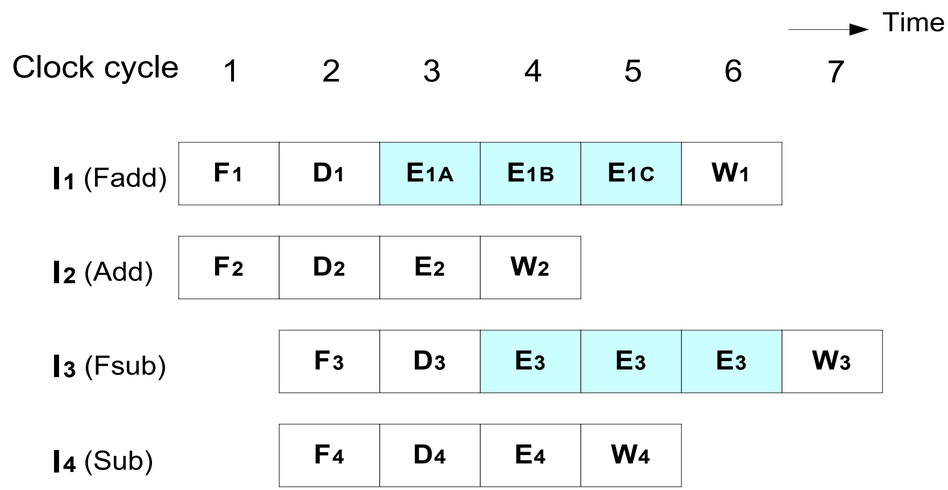
Lecture 9

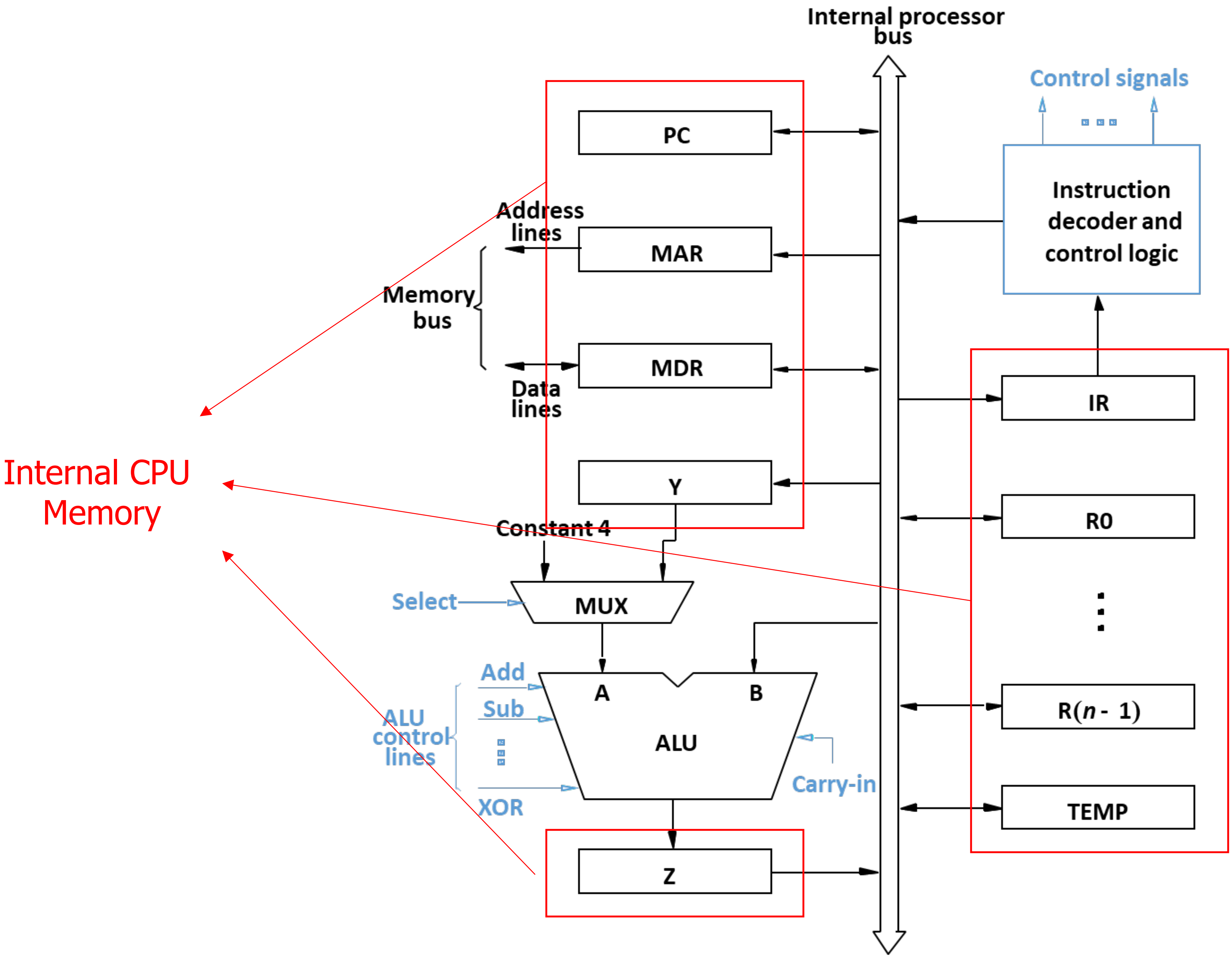
The Memory System

Previously..

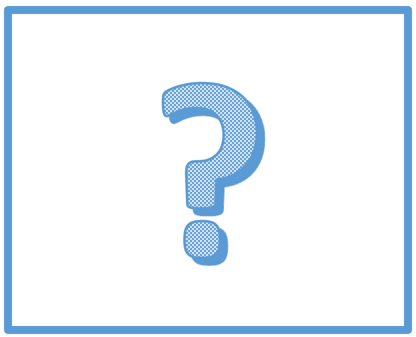
- Pipeline concept
  - Two- and Four- stage pipeline
  - Stage buffer
  - Stall
- Cause of stall (Hazard)
  - Data, Control, and Structural Hazard
  - Operation forwarding
  - Branch prediction
- Superscalar

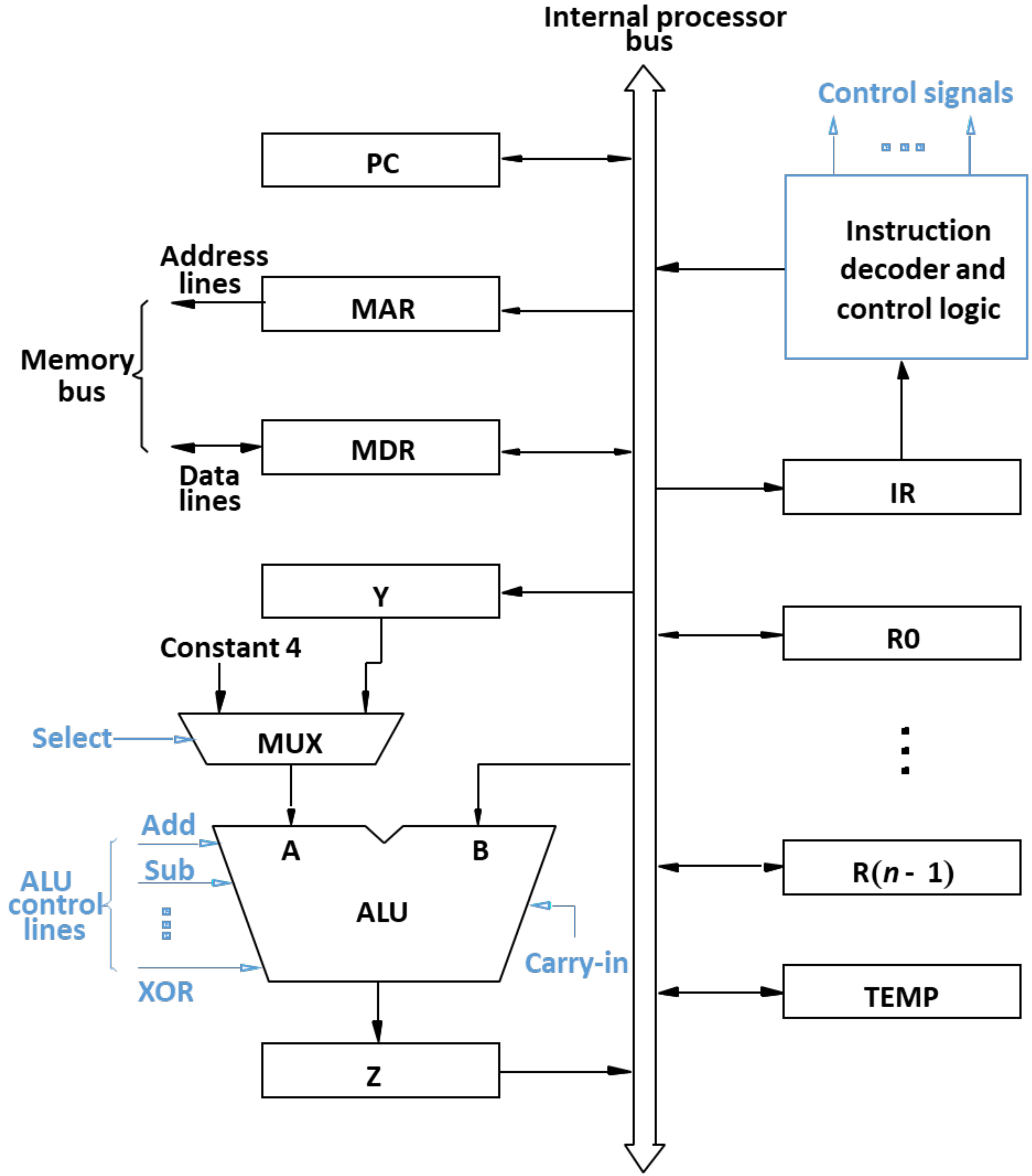




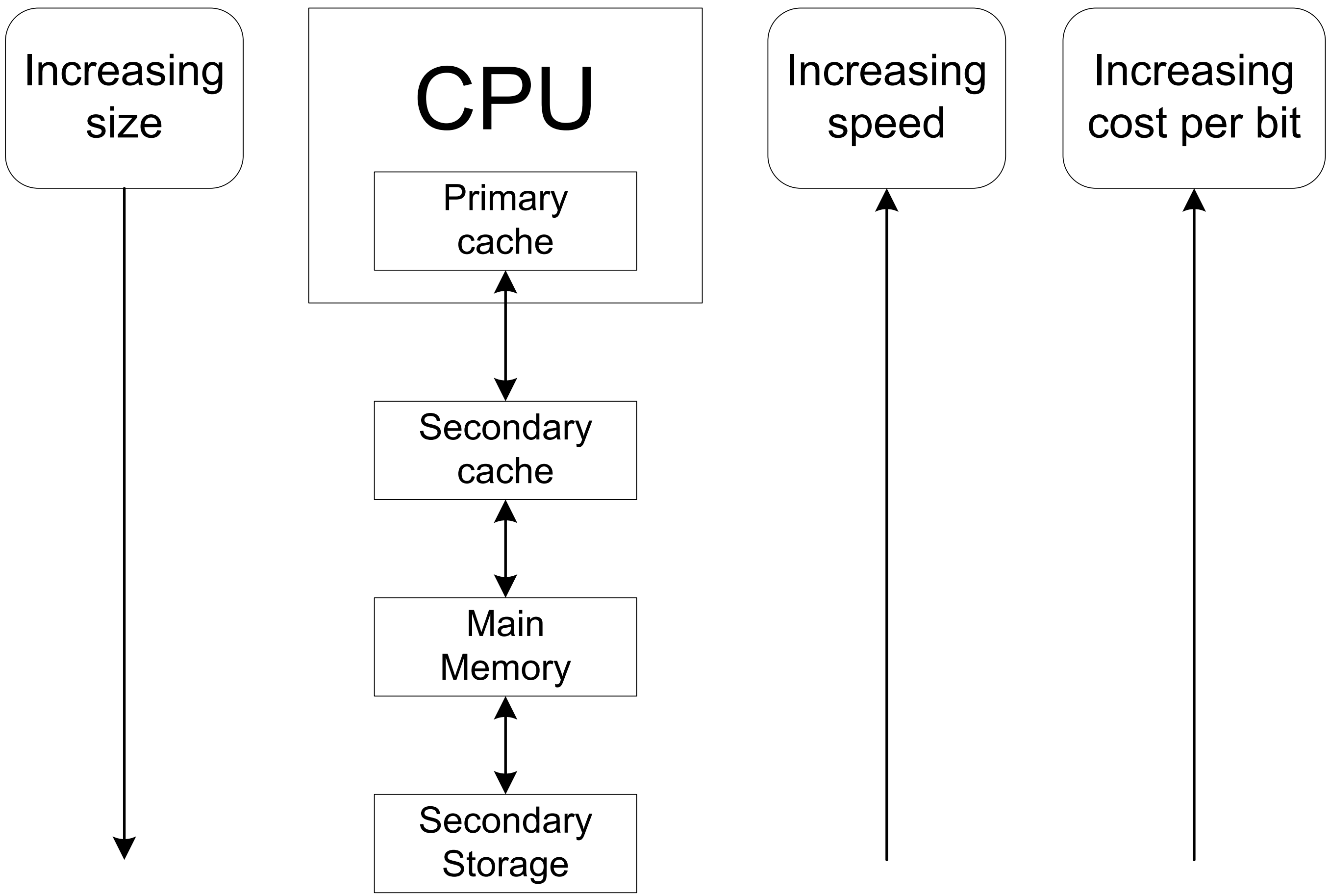


Internal processor bus
Control signals
Instruction decoder and control logic
PC
Address lines
MAR
Memory bus
MDR
Data lines
Y
Constant 4
Select
MUX
IR
R0
R(n - 1)
TEMP
Add
Sub
ALU control lines
XOR
A
B
ALU
Carry-in
Z
Internal CPU Memory




Internal processor bus
Control signals
PC
Address lines
MAR
Memory bus
MDR
Data lines
Instruction decoder and control logic
IR
Y
R0
Constant 4
MUX
Select
Add
Sub
ALU control lines
XOR
A
B
ALU
Carry-in
R(n - 1)
TEMP
Z

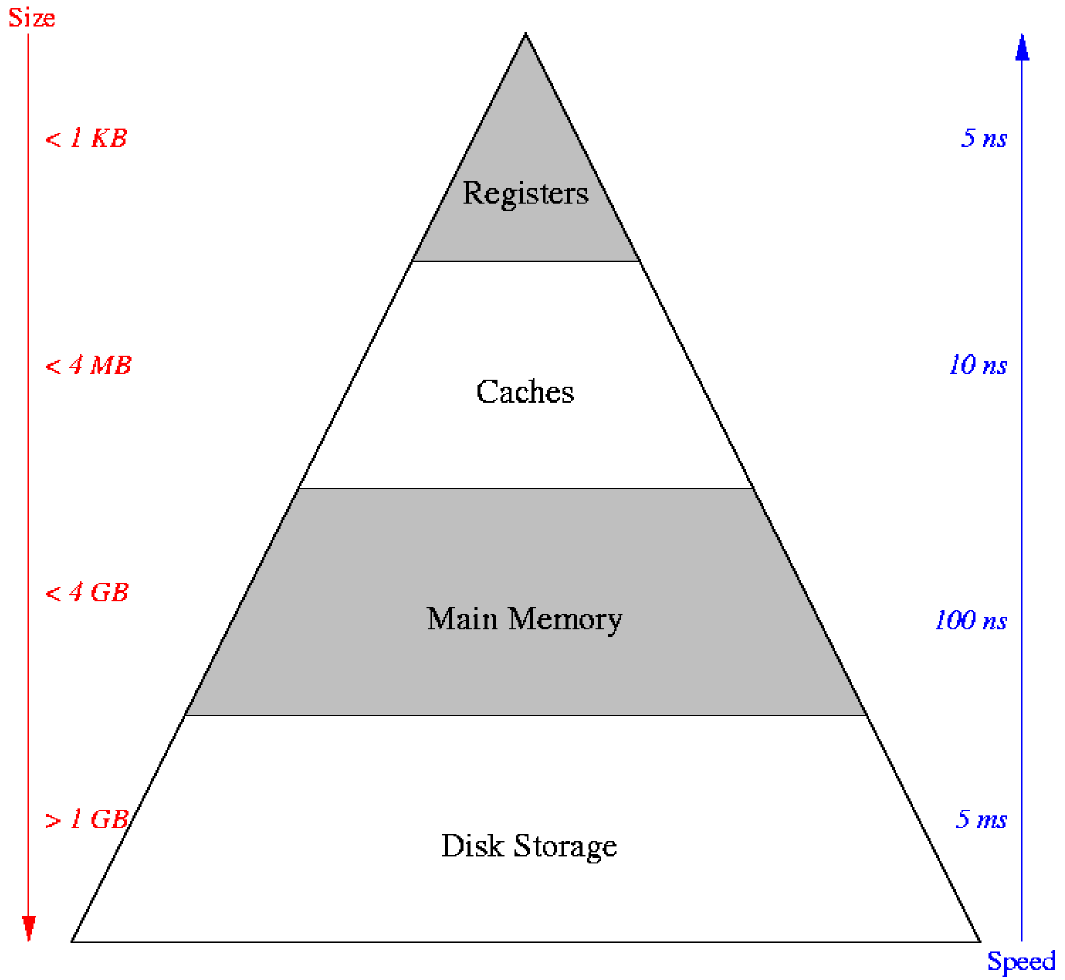
Memory System: Speed, Size, and Cost



The Memory System

- Main memory holds
  - Execution programs (instructions sequence)
  - Data that the programs operate on
- When turn-off the computer, executable programs are stored in the secondary storage in form of executable files.
- The execution speed of the program depends largely on the data transfer speed between the processor and the main memory.
- The memory hierarchy is designed to improve speed while keeping the cost reasonable.

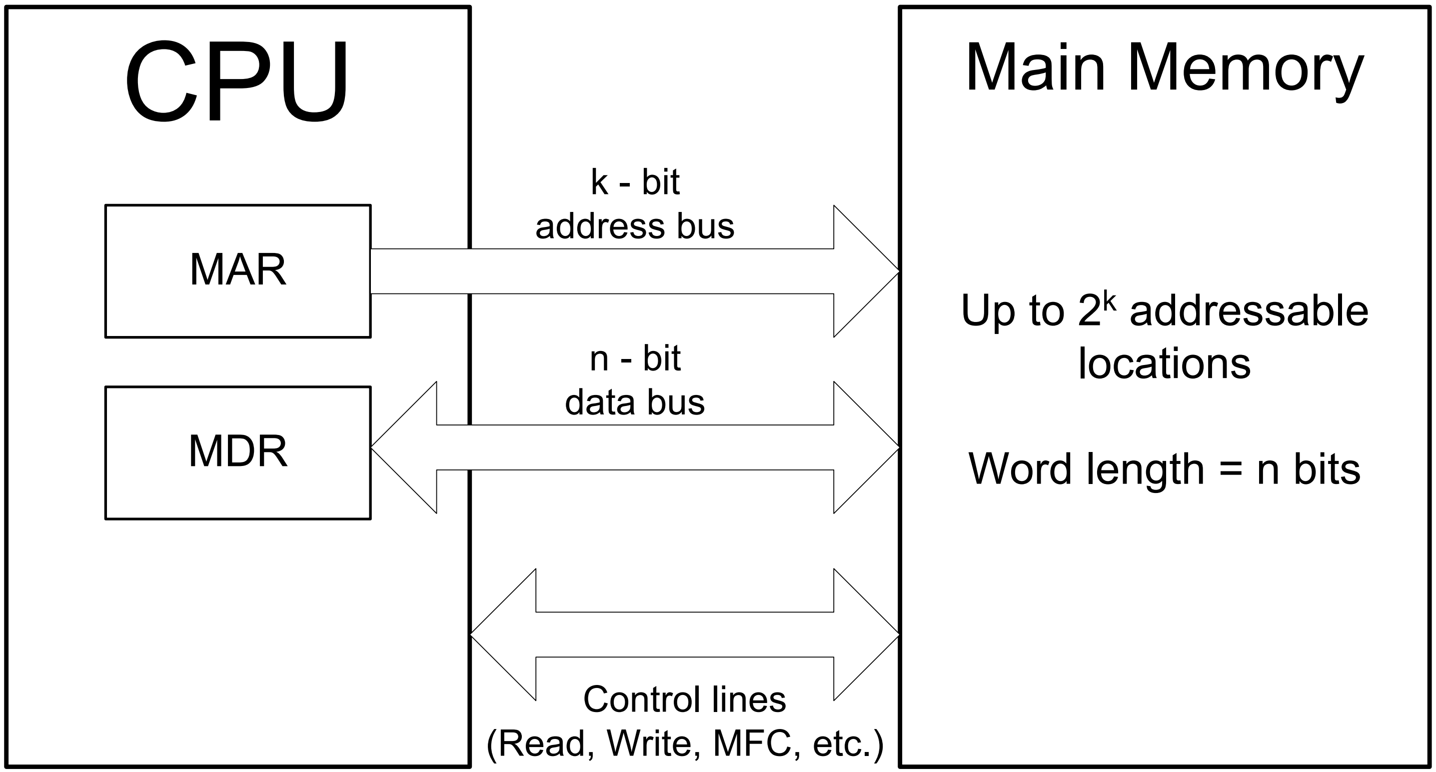
Memory Hierarchy



Main Memory

- Data can be stored to and retrieved from a memory in word size.
  - Word size refers to the number of bits processed by CPU.
  - Data bus size, instruction size, address size are usually multiples of the word size.
- The maximum size of the memory is determined by the addressing scheme. For example,
  - A 16-bit (word) computer (16 bit addresses) is capable of addressing up to 2^{16} = 64KB of memory.
  - A 32-bit (word) computer can handle 2^{32} (4 GB).

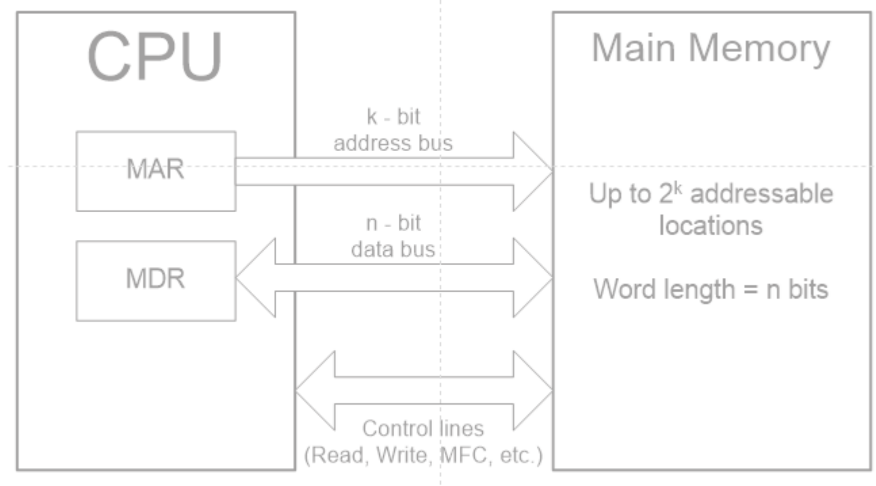
Connection Between Memory and CPU



Data Location and Data Word

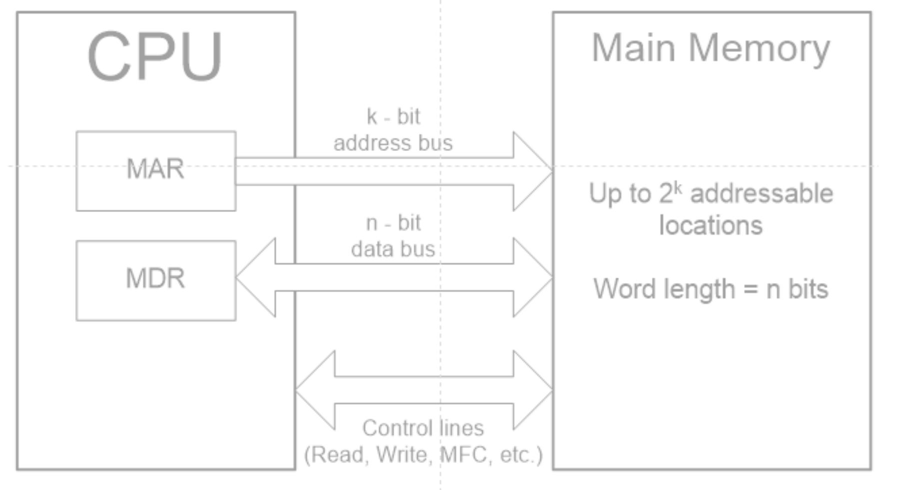
- Data transfer takes place through the use of processor registers.
  - MAR (Memory Address Register) k bit length
  - MDR (Memory Data Register) n bit length
- A memory unit may contain up to 2^k locations and during each memory cycle, *n* bit data can be transferred.
- Data transfer occurs on the processor bus which contains *k* address lines and *n* data lines. The bus includes the control line, such as read and write for coordinating data transfer.

Read from The Memory



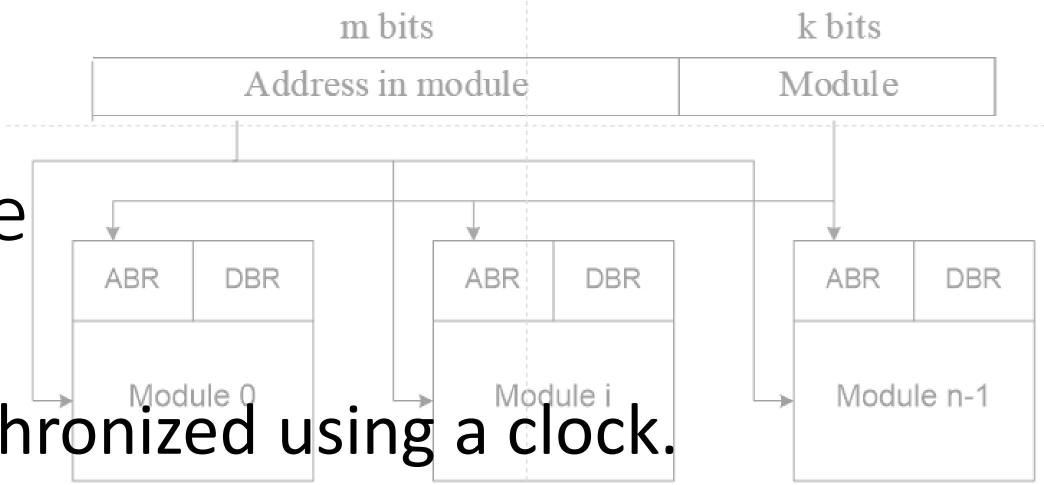
- Data Read: Operation step
  - The processor loads the required address on to the MAR
  - The processor sets R/W to 1
  - The memory places the addressed data on the data line
  - The memory confirms the action by setting the MFC signal
  - When the processor reads the MFC, it loads the data from the line to the MDR

Write to The Memory



- Data Write: Operation step
  - The processor writes to the MAR and the MDR
  - The processor sets R/W to 0
  - The memory reads from both buses

Memory Access Time



- Memory access is synchronized using a clock.
- Memory access time is counted from the initiation of the access instruction to the completion.
- Memory cycle time is the minimum delay required between the initiation of two successive memory operation.
- Memory interleaving can improve memory access time by separating memory in modules.

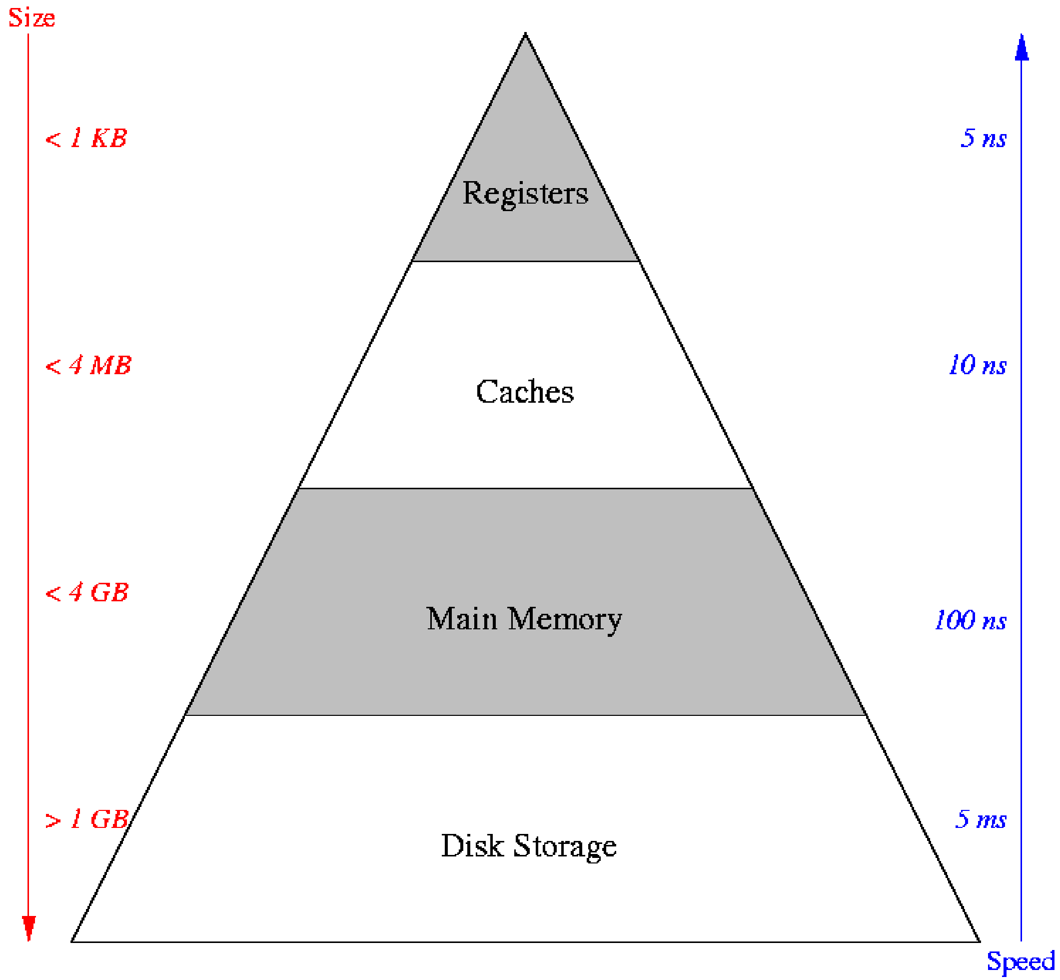
Random Access Memory (RAM)

- RAM is a memory unit where any location can be accessed in the same pattern, **fixed amount of time** independent of address location .
- RAM is different from the serial access storage device such as tape and disk because the access time of a serial storage **depends on the position** of data.

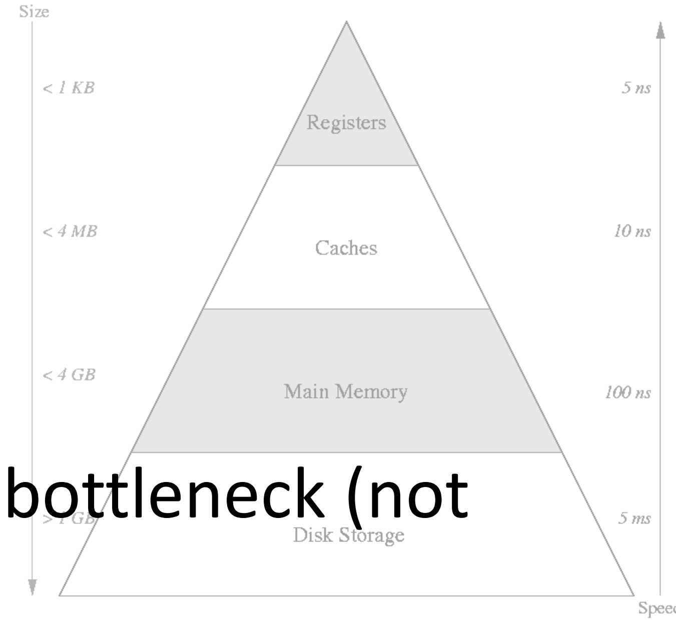
Read Only Memory (ROM)

- ROM is used to implement parts of the main memory that contain fixed programs or data.
- ROM is a non-volatile memory device. The read operations can be performed in the same way as RAM. However, the write operation needs special processes.

Memory Hierarchy

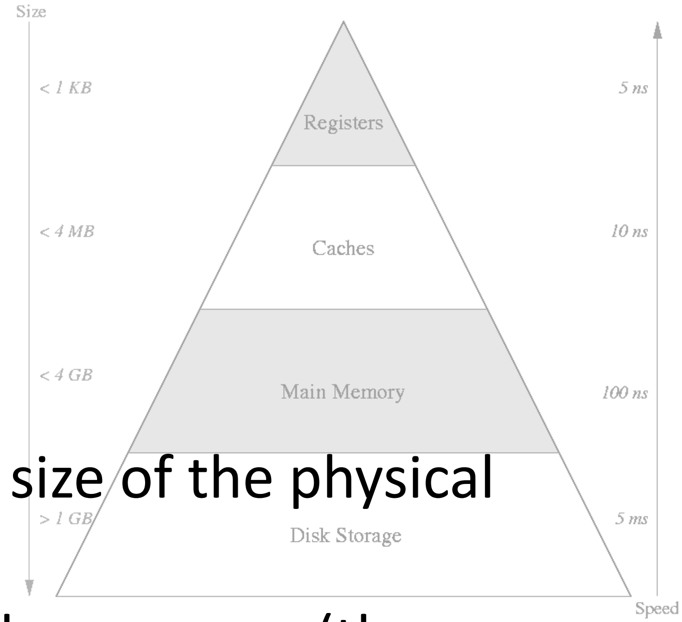


Cache Memory



- Memory cycle time is the system bottleneck (not the instruction processing time).
- Cache is thus introduced. It is a small but fast memory that can be inserted between a processor and a main memory. It is used to hold active segment of a program and data.

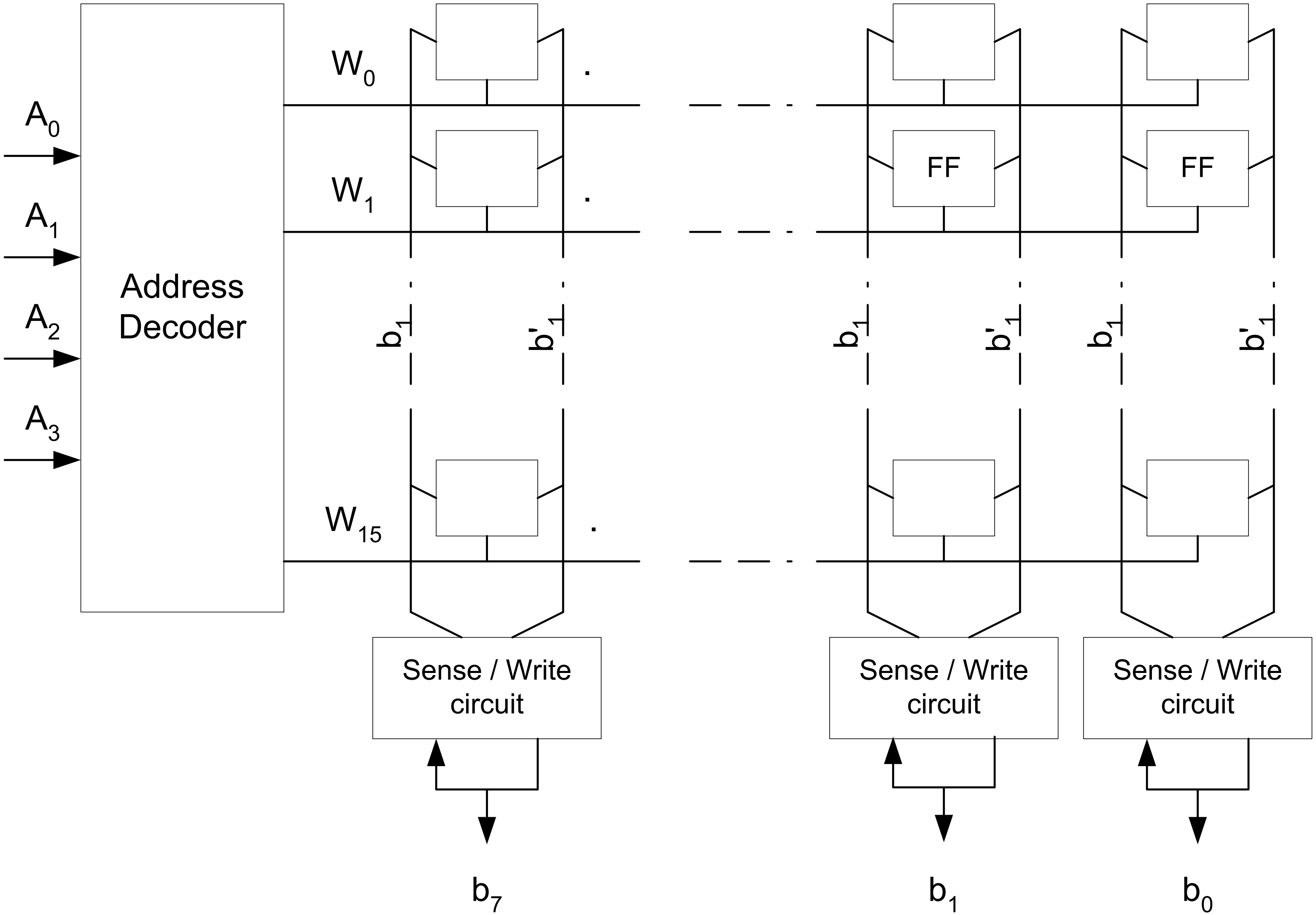
Virtual Memory



- Virtual Memory is used to increase the size of the physical memory
- In many cases, addresses specified by the program (the logical address) are **different from the physical address**. The address translation is achieved by using the memory control circuit.
- The virtual memory is bigger than the physical memory. At any given time, only a part of it is mapped to the actual memory, the rest is mapped to disk.

Random Access Memory (RAM)

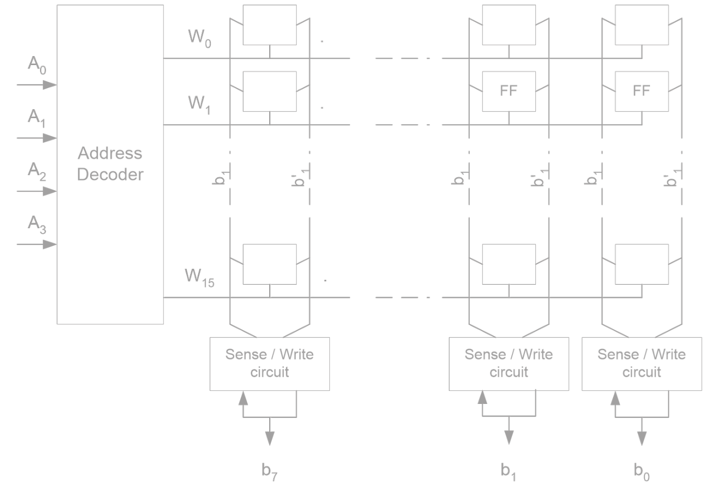
RAM Architecture



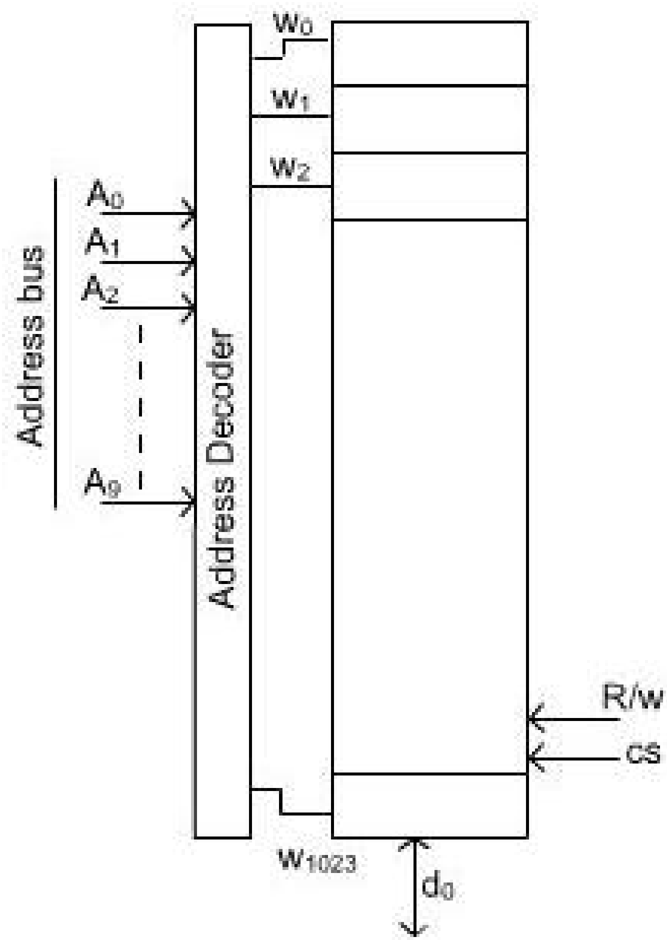
Internal Operation of RAM Chips

- Memory cells are organized in the form of an array. Each cell store 1 bit of information.
- All cells in the same row are connected to a word line, which is connected to the address decoder.
- All cells in the same column are connected to a bit line. The bit line is then connected to the sense (read/write) circuit.
- Read : a circuit reads information from the cells selected by a word line.
- Write : a circuit receives and stores information from and to cells of a selected word.

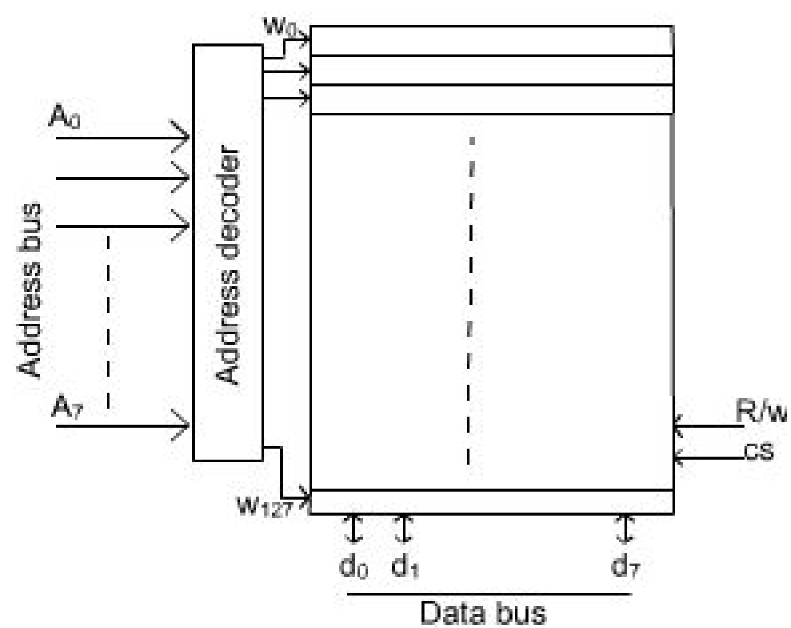
How to Organize RAM



1K bit memory: 1024x1 memory chip

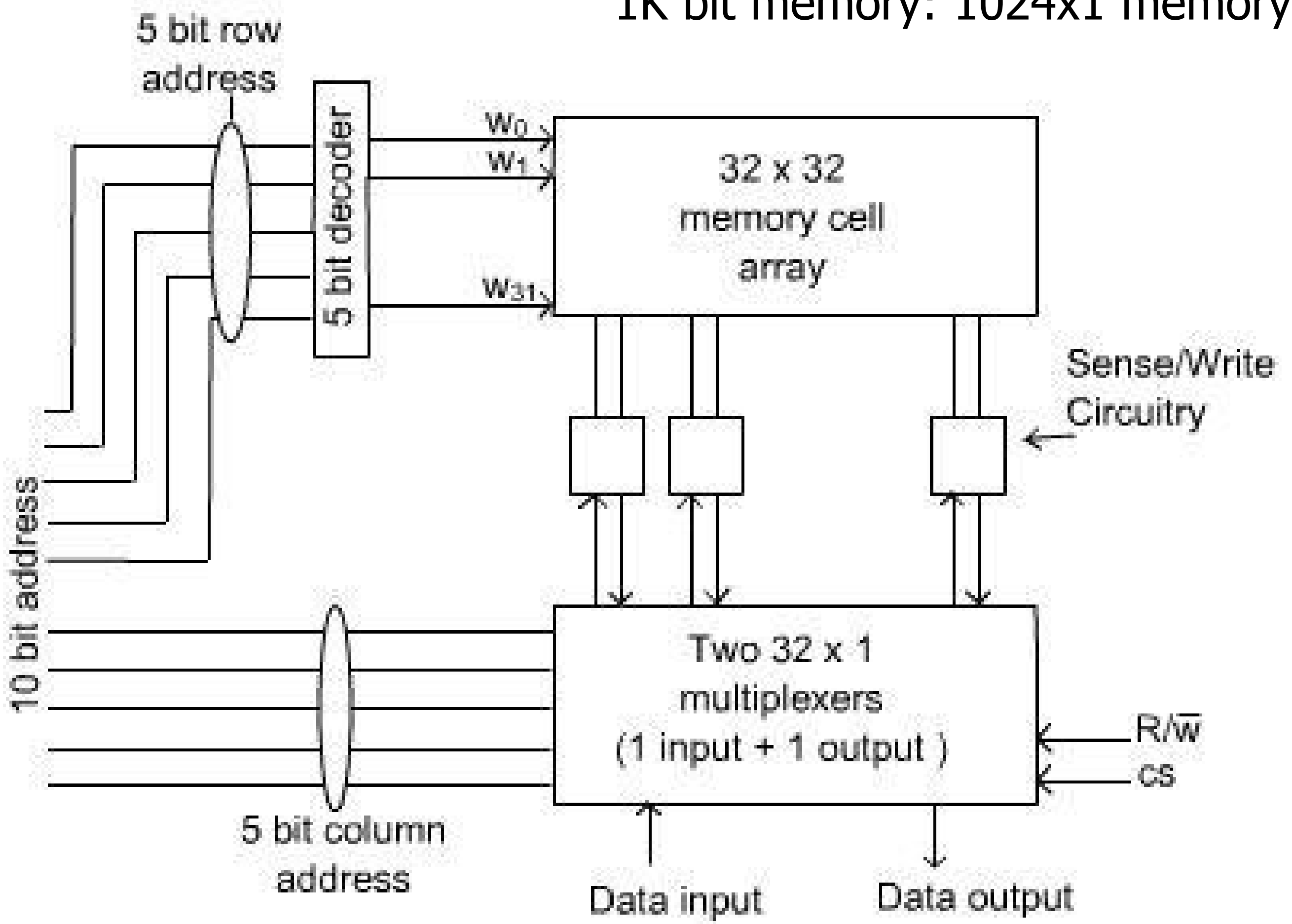


1K bit memory: 128x8 memory chip

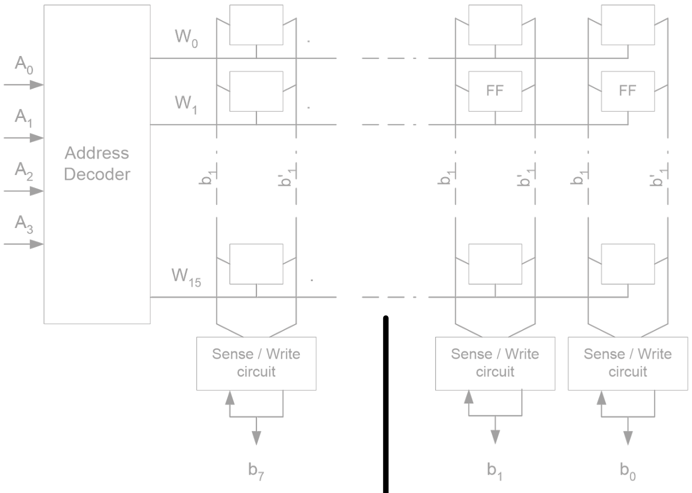


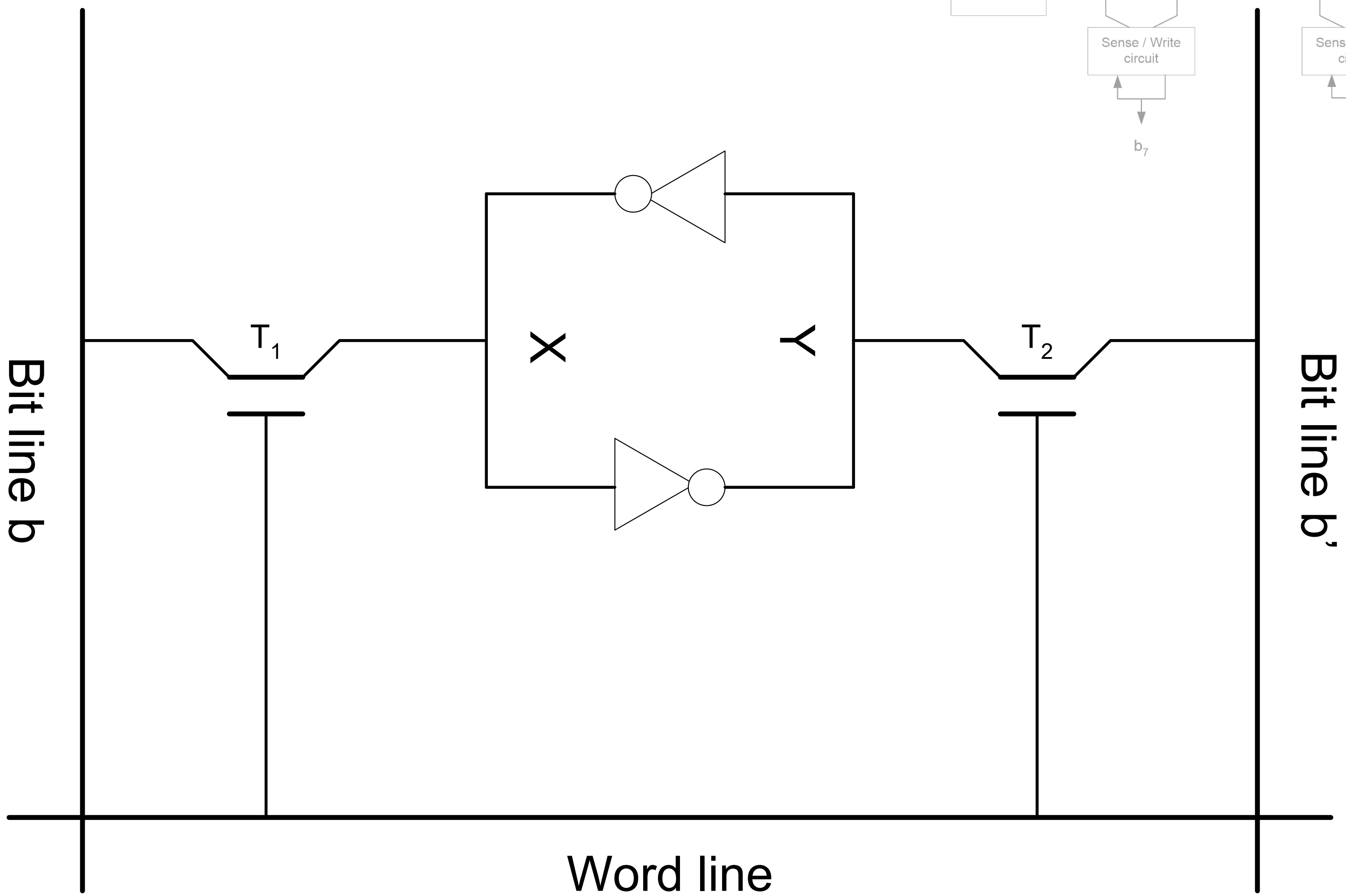
Sample of RAM Organization

1K bit memory: 1024x1 memory chip



SRAM Circuit

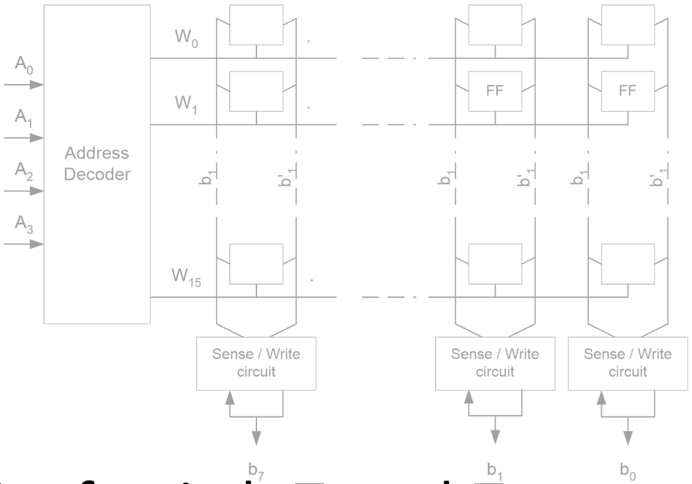




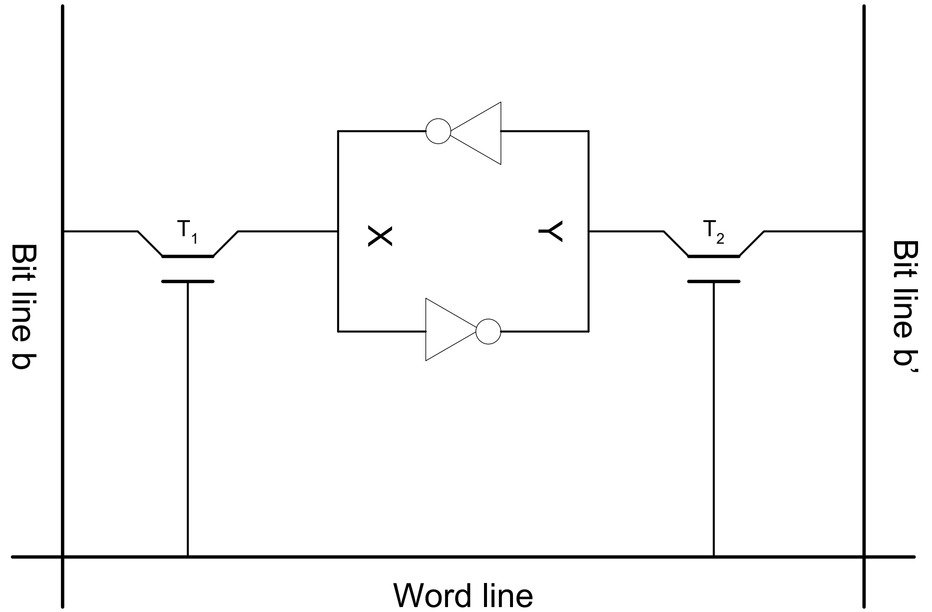
Static RAM (SRAM)

- Each RAM cell consists of 2 inverters cross-connected to form a latch.
- A latch is connected to bit lines through transistors.
- If x is 1 and y is 0, the cell state is 1.
- As long as a signal on the word line is grounded, the cell will retain the state.
- When the power is interrupted, states in SRAM will be lost.

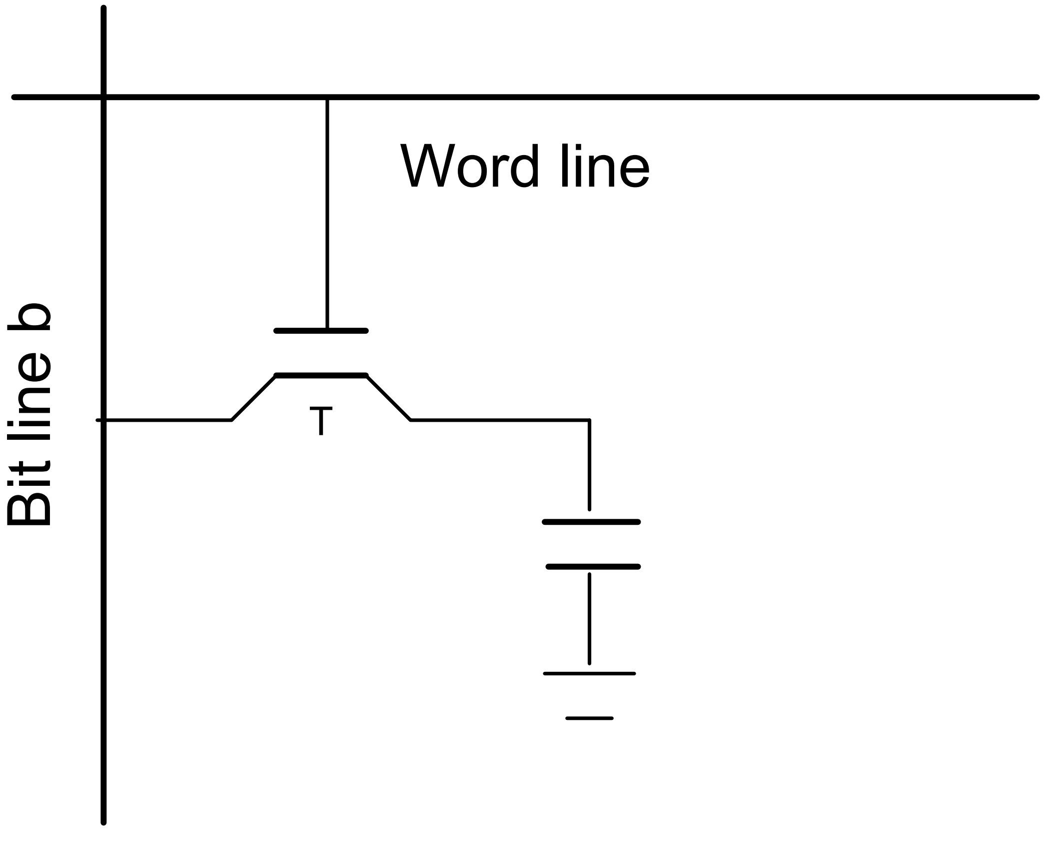
SRAM Operation



- Read Operation
  - A word line is activated to close circuit of switch T_1 and T_2.
  - If *b* is high and *b'* is low, the state is 1.
  - If *b* is low and *b'* is high, the state is 0.
- Write Operation
  - Place the correct state on *b* and *b'*, and activate the word line.



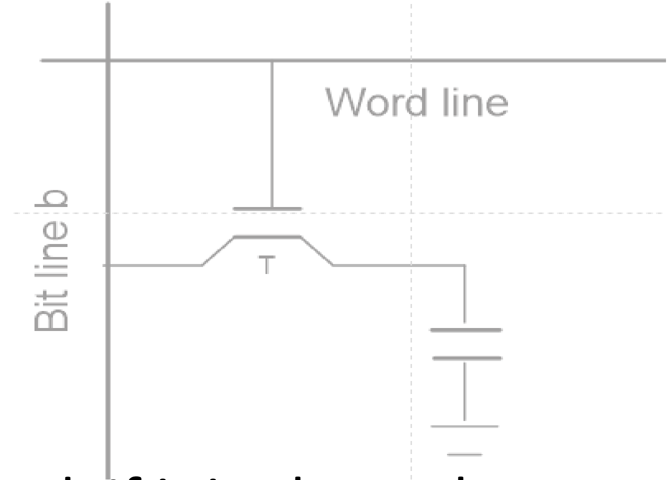
DRAM Circuit



(Asynchronous) Dynamic RAM (DRAM)

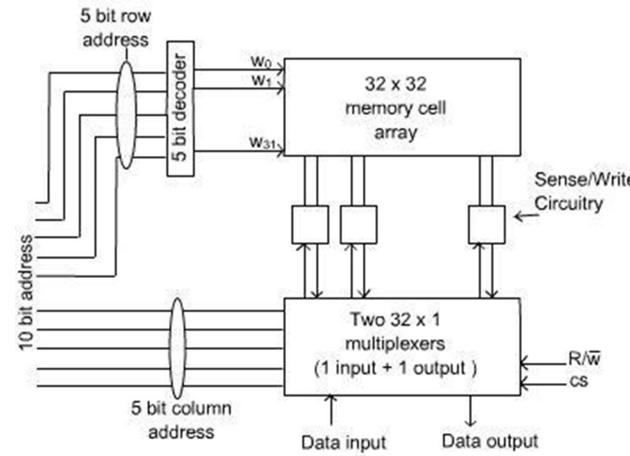
- SRAM is fast, but expensive with several transistors. DRAM circuit is simpler. However, it cannot retain the state indefinitely.
- Information stored in DRAM is in the form of a charge on a capacitor. The contents must be periodically recharged (The charge can only be held for some 10 milliseconds).
- To store information in the cell, a transistor *T* is turned on, which causes a known amount of charge to be stored in the capacitor.

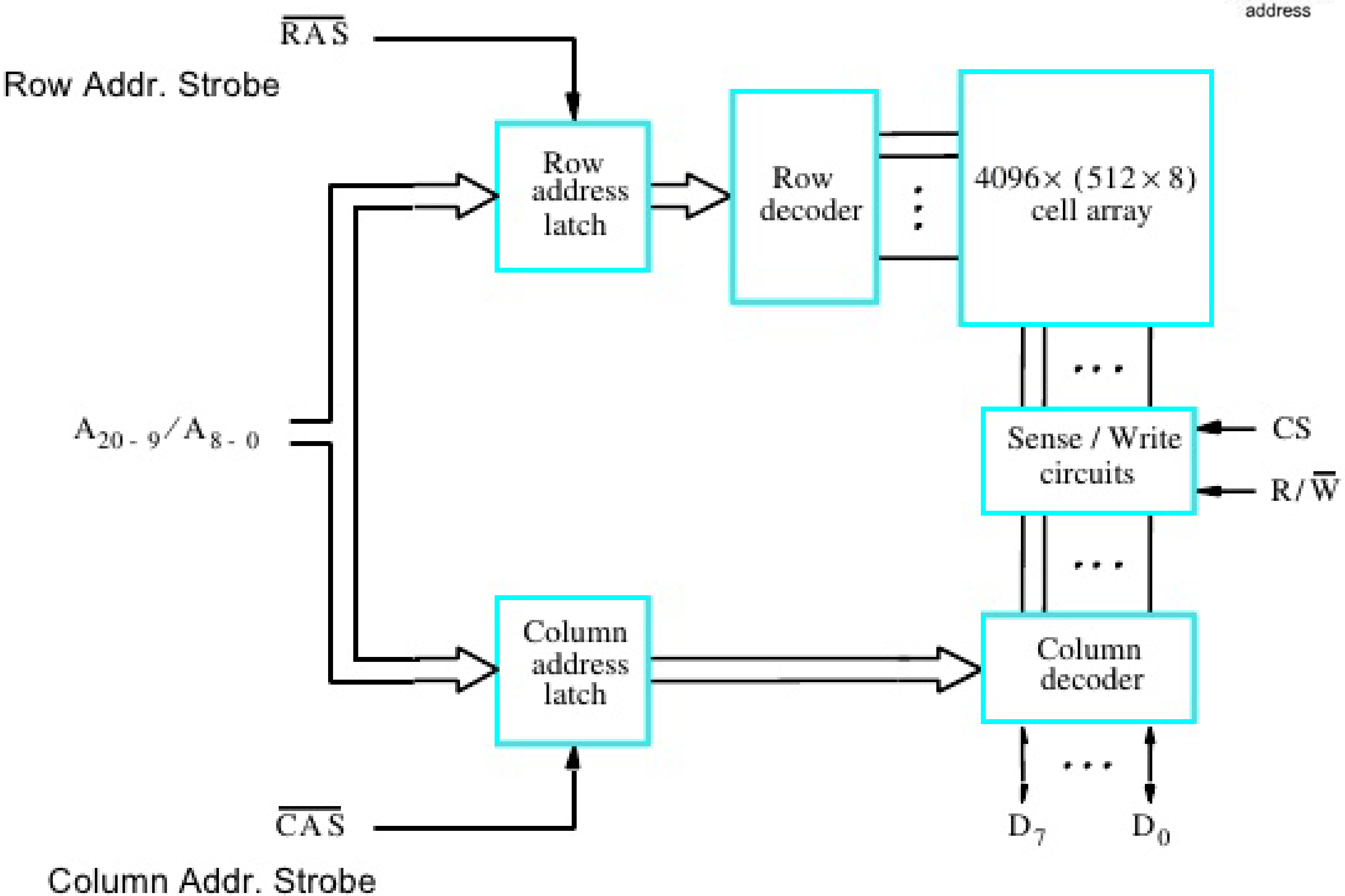
DRAM Operation (1)



- Read Operation
  - the charge in the capacitor is measured. If it is above the threshold value, the bit line is drove to full voltage to represent value 1. As a result that the capacitor is recharged.
  - if the charge is below the threshold, the bit line is pulled to ground for value 0, the capacitor is drained.
  - Reading the cell, thus causes the auto-recharge to the capacitor.

DRAM Organization





16M bit memory: 2Mx8 memory chip

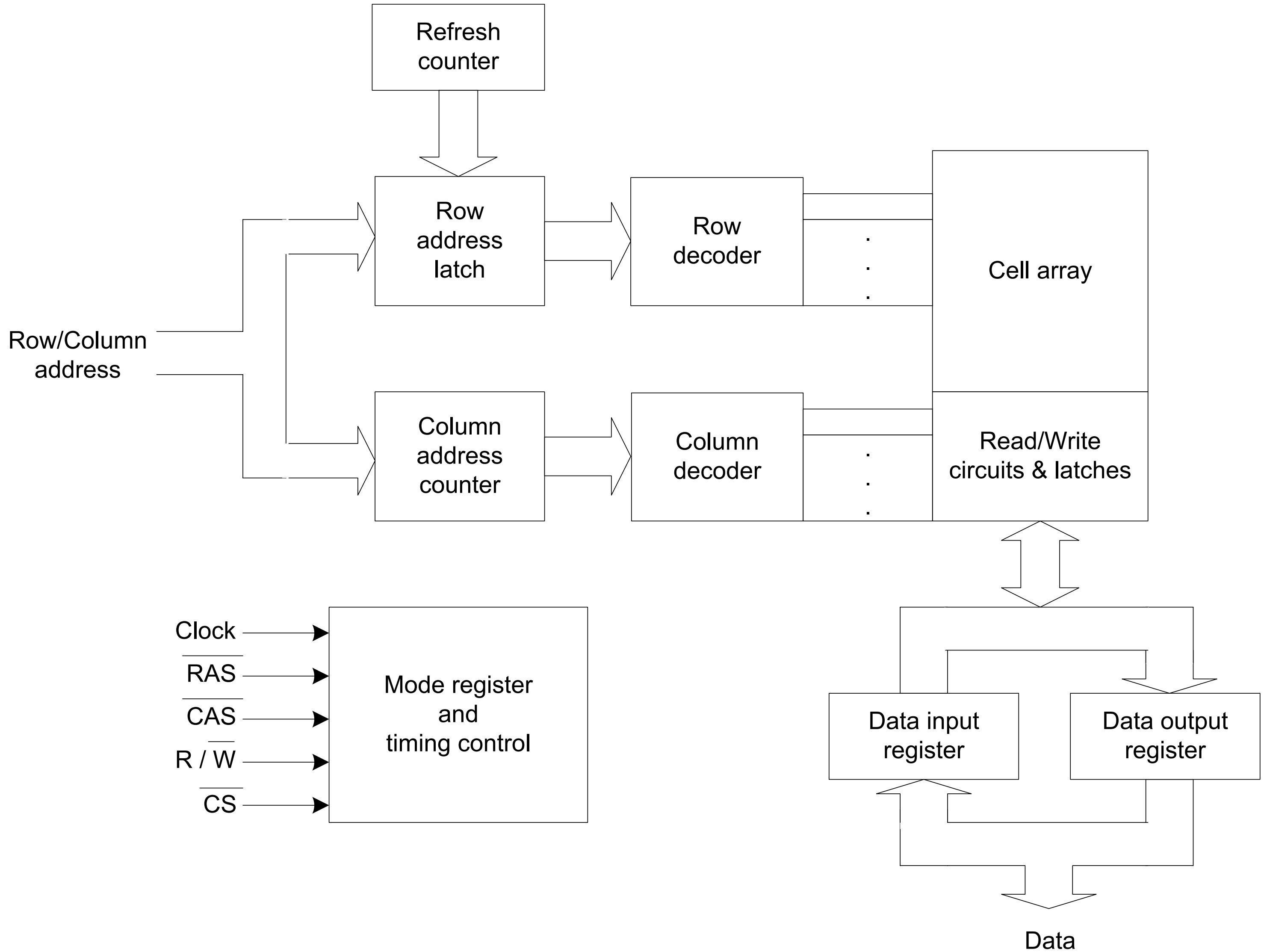
DRAM Organization (2)

- The cell array of the DRAM in the figure above has the size of *4k*4k bit*, called a 2M × 8 dynamic memory chip.
  - 16 M-bit, Each row can store 512 bytes of data (1 byte = 8 bits). There are 4096 rows in the memory.
  - 12 address bits are used to select a row and an extra 9 bits are needed to select a group of 8 bits in the selected row.
  - The total of 21 address bits are needed.
  - The timing of the memory is controlled asynchronously. RAS (Row Address Strobe) and CAS are used as control signals to govern the timing.

DRAM Organization (3)

- During read/write operations, row address is applied first, then the operation is initiated, in which all cells on the selected row are read and refreshed. After the row is loaded, the column address is applied.
- An appropriate group of 8 bits is then selected. If the R/W signal indicates read, data of selected group of 8 cells are put on $D_7..D_0$. if indicates write, data from $D_7..D_0$ are put in the cells
- When a byte is fetched, all *4k* cells in the rows are sensed but only 8 bits are read/write.
- Apply row address causes the cells in the row to be refreshed.

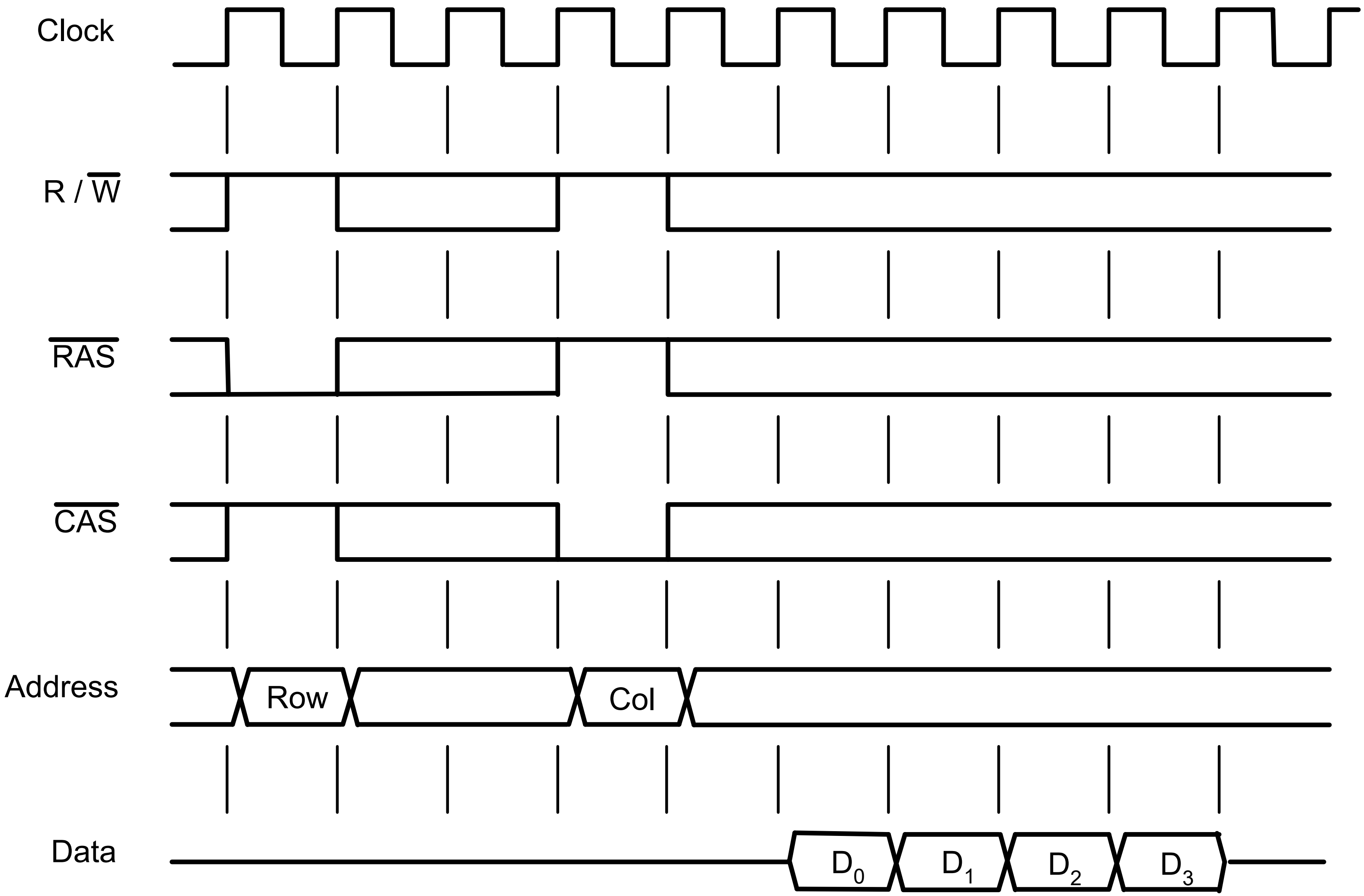
SDRAM Architecture



Synchronous Dynamic RAM (SDRAM)

- SDRAM operation is synchronized with a clock. All data can be placed on the data lines in each clock cycle. All memory actions are triggered by the **rising edge of the clock**.
- SDRAMs have built-in **refresh circuitry**, with a refresh counter to provide the addresses of the rows to be selected for refreshing.
- Unlike asynchronous DRAM, there is no need to externally generate pulse on the CAS to select a column. The control signals are provided internally using **a column counter and the clock signal**.

SDRAM Timing



SDRAM Operation

- Read operation (length 4)
  - RAS is latched to read the selected row.
  - After 2 cycles delay, the CAS is latched. (maybe 3 cycles)
  - An extra 1 cycle delay, the data is placed on the data line one byte per cycle.
  - New data are placed on the data lines **at the rising edge of each clock pulse.**
- In SDRAM, each row must be refreshed at least every 64 ms.
- The initial commercial SDRAMs designed for clock speeds of up to **133 MHz**.

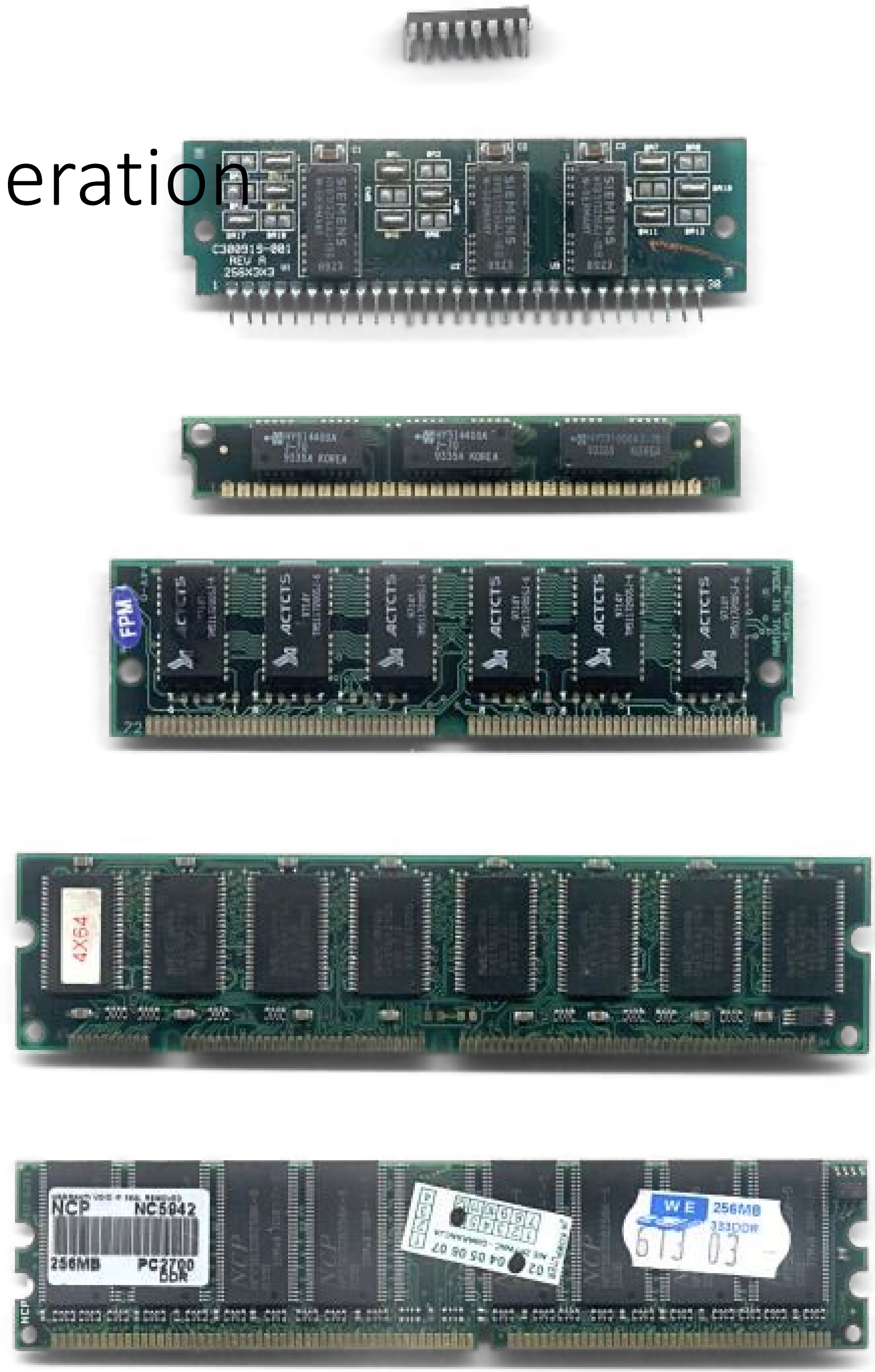
SDRAM Latency

- Memory latency is an amount of time it takes to transfer one word to or from memory. This number indicates memory performance.
- In block transfer, the time to complete operation also includes the delay between successive word transfer.
- Latency in block transfer is usually referred to as the time to fetch the first word. (time includes RAS and CAS setting).
- From the timing diagram above, the first word take 5 cycles and the latter words take 1 cycle each. **The latency in this case is five clock cycles**. If the clock rate is 100 MHz, then the latency is 50 nsec.

Double-data-rate SDRAM (DDR-RAM)

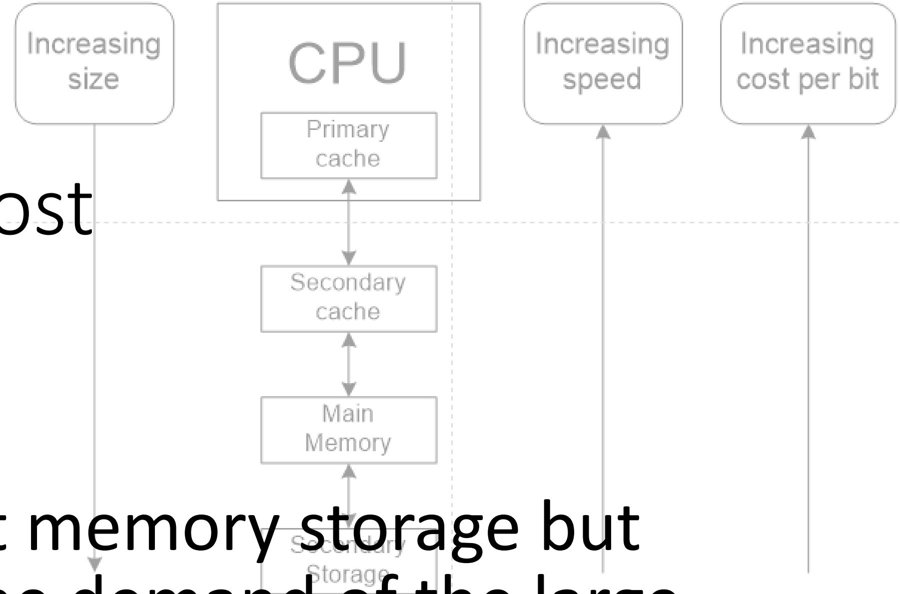
- Data transfer can be done on **both edge of the clock**. The latency of the device is the same as the SDRAM, but the bandwidth for a long burst transfer **can be doubled**.
- Cell array can also be organized into 2 memory banks. Each bank can be accessed separately. Consecutive words in a block can be stored in different banks. Interleaving of words allow access for 2 words/clock cycle, in this case.

Memory System Consideration



- SRAM should be used when fast operation is the primary requirement. It is mostly used for cache memory.
- DRAM is usually used for the main memory due to its high density and inexpensiveness.

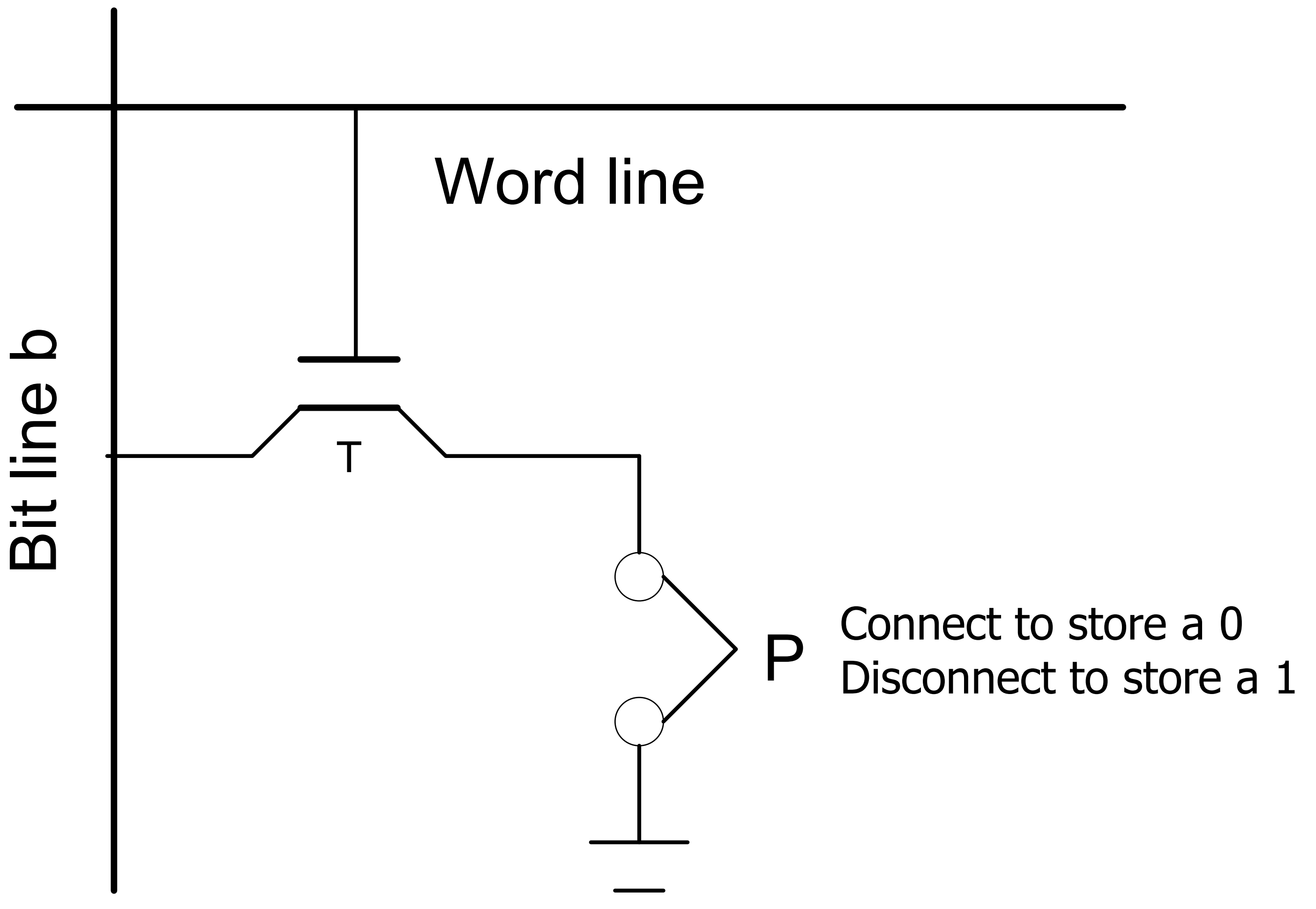
Note: Speed, Size, and Cost



- SRAM is fast but expensive.
- DRAM can be used in a vast memory storage but still too small compare to the demand of the large program.
- A smaller unit with a fast accessing speed, such as cache can be built using SRAM.
- A big and cost-effective storage can be provided by magnetic disk.

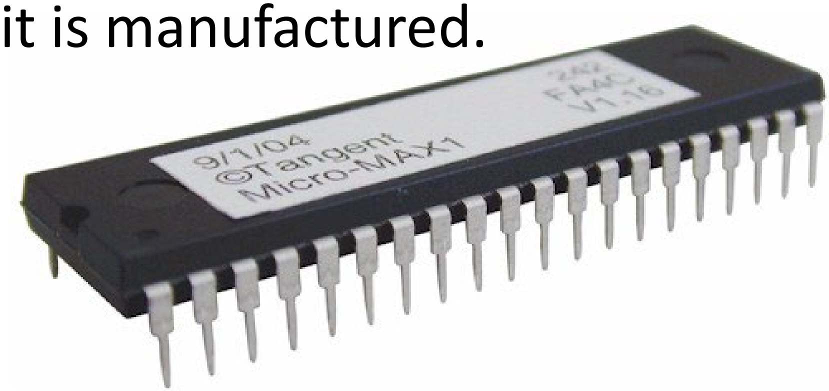
Read Only Memory (ROM)

ROM Circuit


Word line
Bit line b
T
P
Connect to store a 0
Disconnect to store a 1

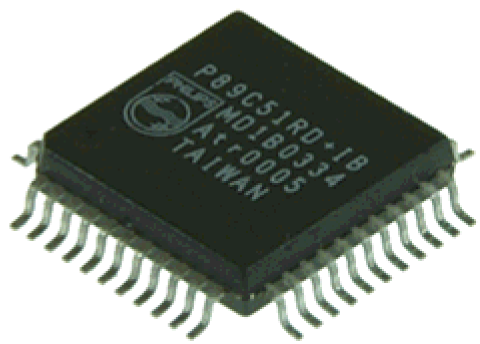
ROM Operation

- Value 0, is stored to the ROM cell if the transistor is connected to ground at point *P*.
- The value of the cell will be 1, otherwise.
- In reading operation, the word line is activated, and the transistor is closed. If the transistor is grounded, the voltage on the bit line will be dropped to near 0 (indicating value 0). Otherwise, the bit line voltage will remain high (indicating value 1).
- Data are written to ROM when it is manufactured.



Programmable ROM (PROM)

- PROM : the data can de loaded by the user. This is achieved by inserting a fuse to the circuit. All cells start with the value 0. The user can, then insert 1 at the required cells by **burning out the fuses**. The process is thus irreversible.
- OTP: One Time Programmable

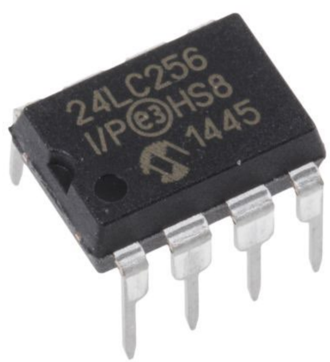


Erasable Programmable ROM (EPROM)

- EPROM : point *P* is always connected to ground and a special transistor is used. The transistor can be programmed to behave like a normal transistor or like an open switch (disabled). This is achieved by injecting charges into the transistor and let them trapped inside. Re-programming can be done by exposing the chip to **the UV light**. This will cause the charges to be dissipated.

Other ROM technology

- EEPROM (Electrical EPROM) : date can be selectively cleared without having to remove the chip from the circuit. The different voltages are needed for erasing, writing and reading.
- Flash memory : technology is similar to EEPROM. The devices, however, has a greater density, a higher capacity and a lower cost per bit. They are popular for use with the handheld device such as MP3 player, digital camera and PDA.

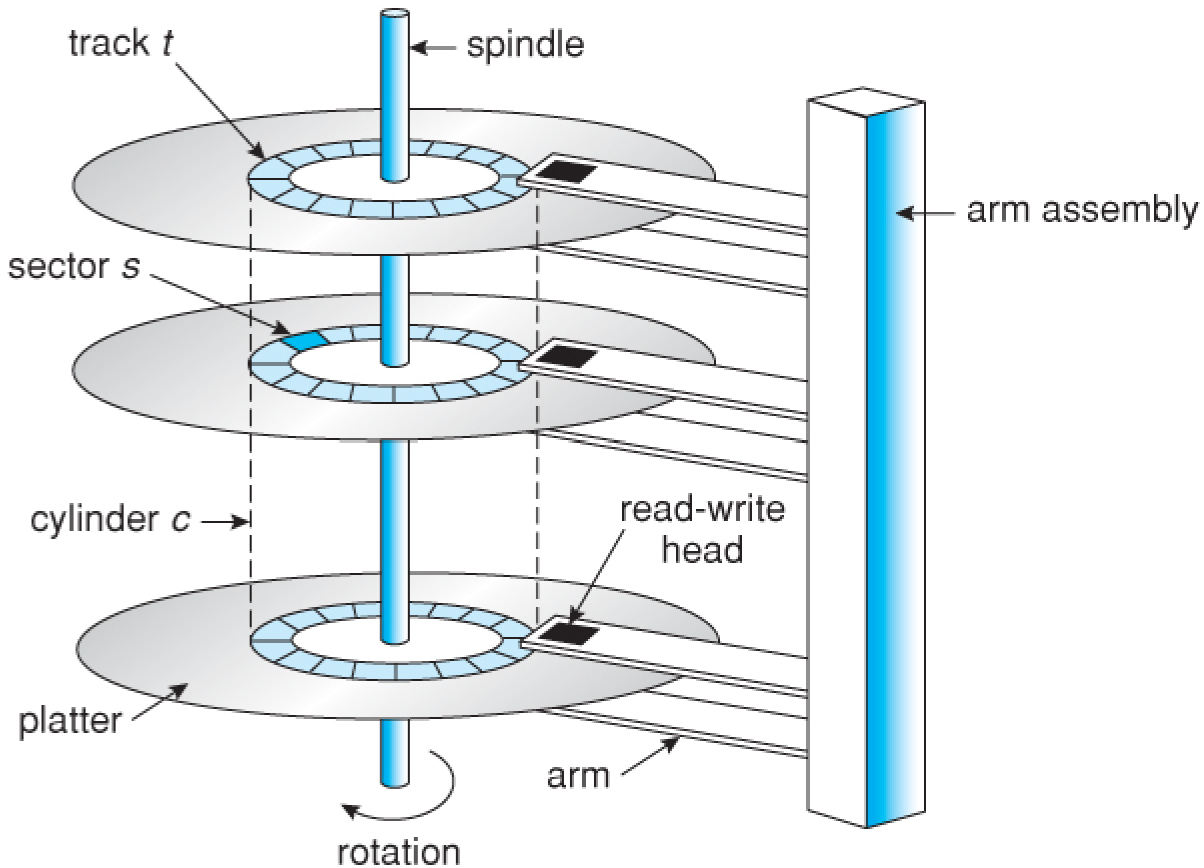


Secondary Storage

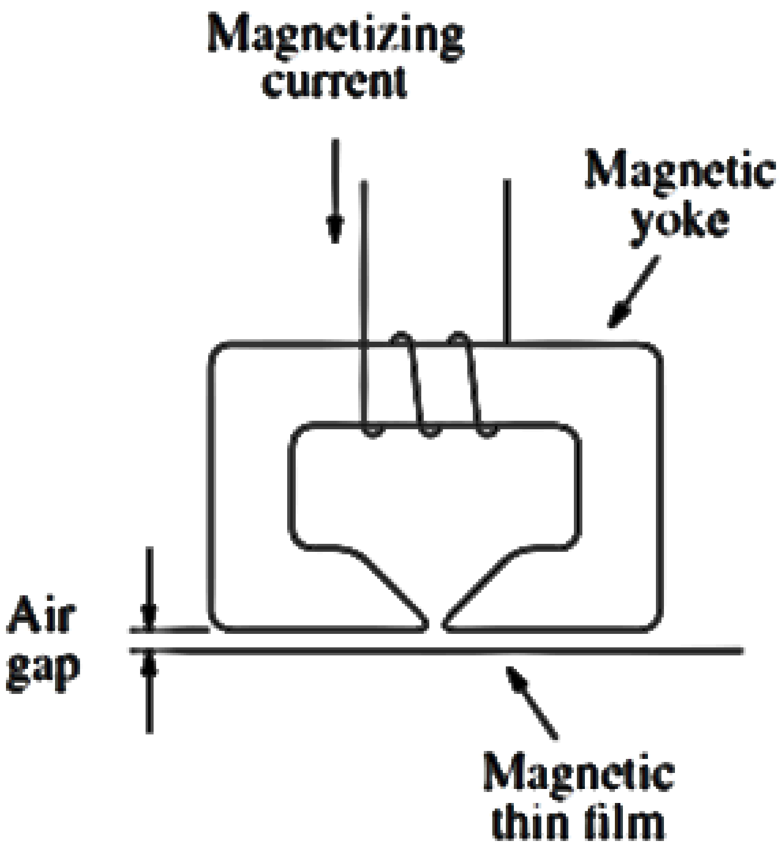
Magnetic Hard Disk

- Disk platter is a flat circular hard surface on which data is stored by magnetic changes on both sides, each of which is called a Surface
- Spindle connects all disk platters together and it is rotated by motor whose rate is measured in rotations per minute (RPM)
- Head is attached with arm above each surface – that arm will move around the surface
- Track is a sub-circle of each surface and each track is divided into sectors – corresponding tracks on each surface form a Cylinder

Disk Geometry



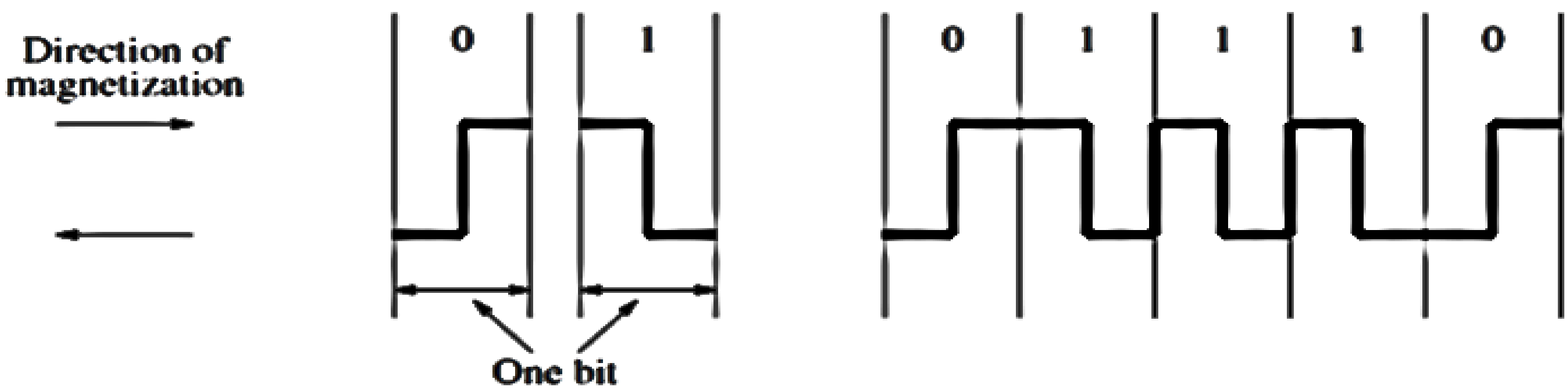
Read/Write Head



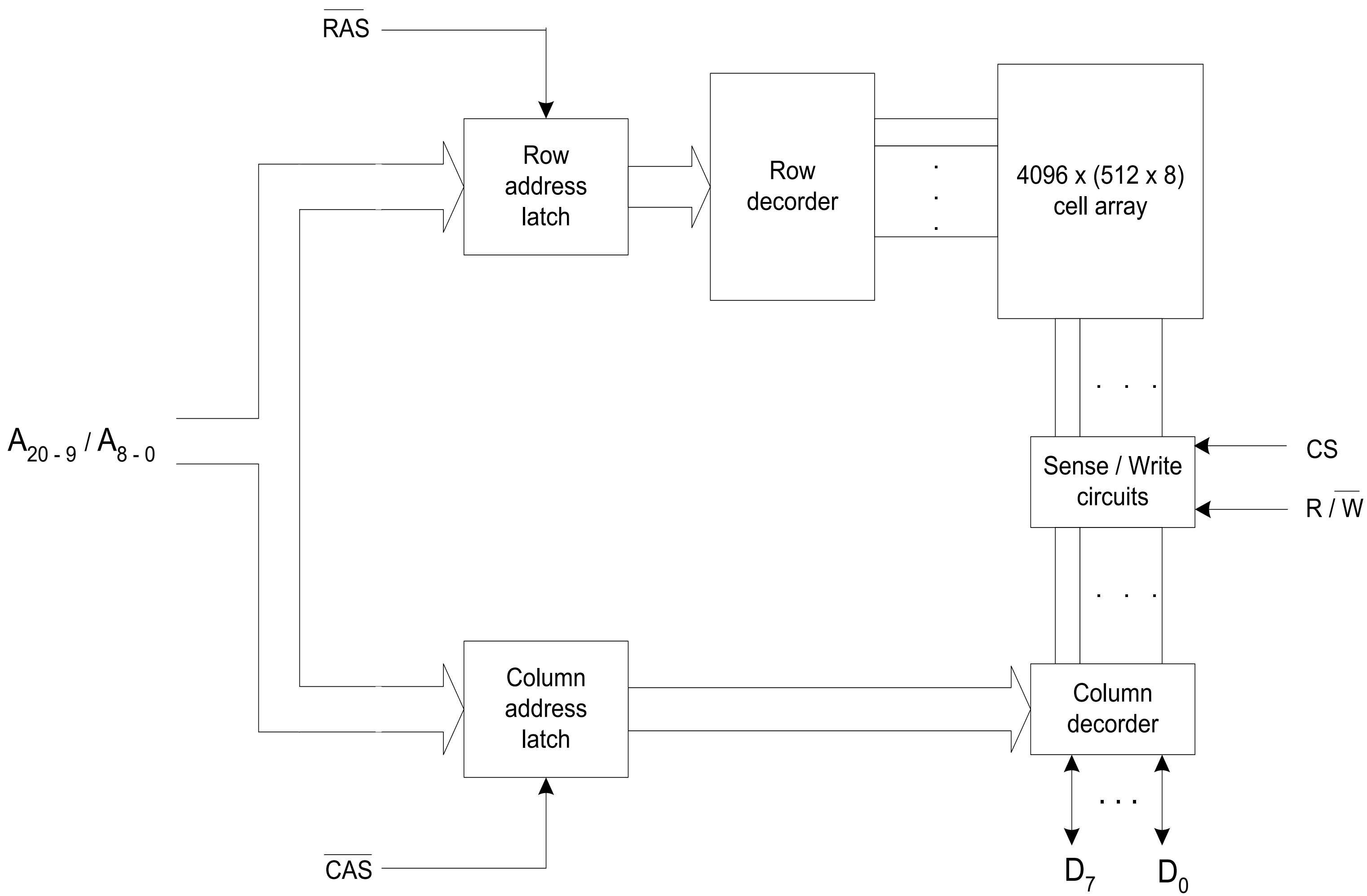
- Each read/write head consists of a magnetic yoke and a magnetizing coil.
- Digital information can be stored on the magnetic film by applying current pulses of suitable polarity to the magnetizing coil.
- This causes the magnetization of the film in the area immediately underneath the head to switch to a direction parallel to the applied field.
- For reading the stored information the magnetic field in the vicinity of the head caused by the movement of the film relative to the yoke induce a voltage in the coil, which now serves as a sense coil.

Data Encoding

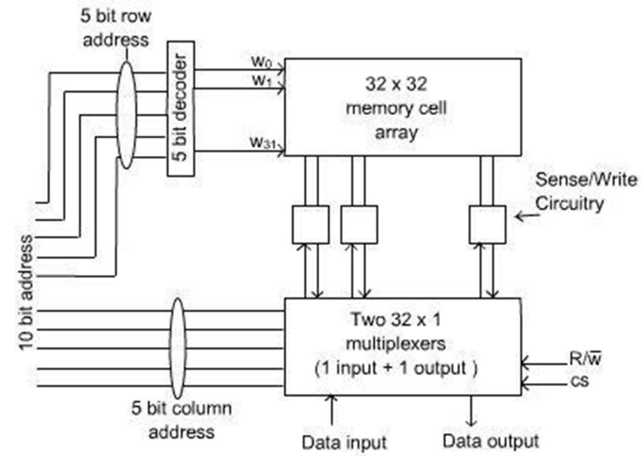
- Several different techniques have been developed for encoding. One simple scheme is p*hase encoding* or *Manchester encoding*.
- Clocking information is provided by the change in magnetization at the midpoint of each bit period.
- The drawback of Manchester encoding is its poor bit-storage density.
- The space required to represent each bit must be large enough to accommodate two changes in magnetization.

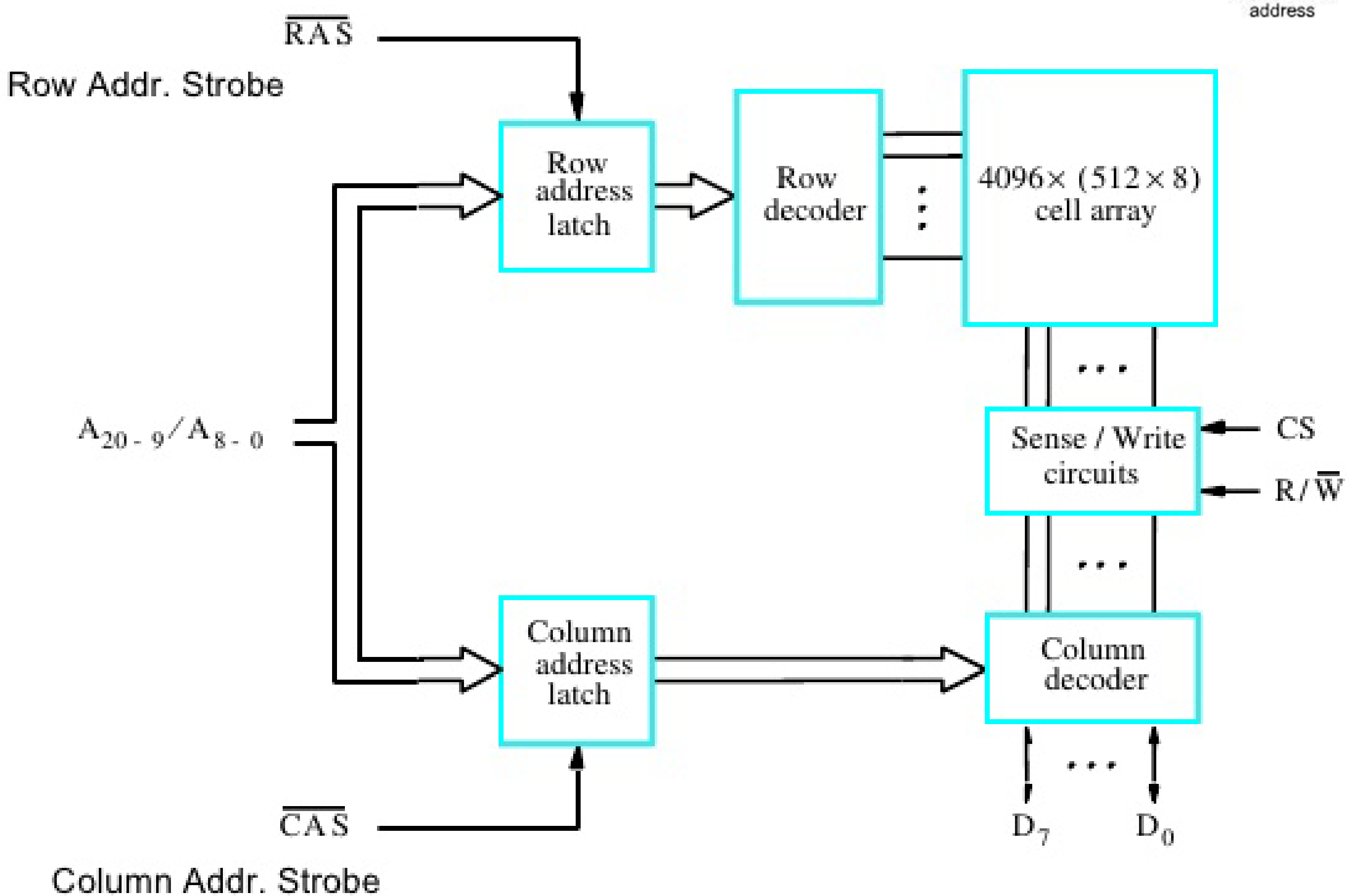


Supplementary Section


$\overline{RAS}$
Row address latch
Row decorder
4096 x (512 x 8) cell array
A_{20-9} / A_{8-0}
Sense / Write circuits
CS
$R / \overline{W}$
Column address latch
Column decorder
$\overline{CAS}$
D_7
D_0

SRAM Organization





16M bit memory: 2Mx8 memory chip